

Compact high-efficiency DC-DC buck converter module. LM2596S-5.0 regulator, 33 μ H inductor and SS54 Schottky diode to provide a stable 5V output.

1 Features

- Input voltage range from 7V to 25V DC
- Fixed 5V regulated output using LM2596S-5.0
- High efficiency (up to 80%) with low output ripple
- Output current up to 2A continuous, 3A peak
- Schottky diode (SS54) for fast switching and low power loss
- 33 μ H power inductor for stable current regulation
- Multiple 220 μ F electrolytic capacitors for input and output filtering
- On-board LED indicator for output status
- Screw terminal connectors (KF128-5.0-2P) for easy input and output wiring
- Compact PCB layout, suitable for integration in embedded and industrial applications
- Thermal and short-circuit protection inherent to the LM2596 design

2 Applications

- Automation and control modules operation
- Industrial equipment requiring a regulated 5V power supply
- Battery-powered devices and mobile electronic systems
- DIY electronics and prototyping projects requiring a compact, efficient power module
- Educational demonstrations involving DC-DC conversion and power electronics

3 Description

The DC-DC Buck Converter Module is a compact power supply designed to provide a stable 5V regulated output from a wide range of DC input voltages.

At its core, the module integrates the LM2596S-5.0 monolithic step-down regulator, which delivers high efficiency and reliable performance for a variety of embedded and industrial applications.

The circuit includes a Schottky rectifier diode (SS54) and a 33 μ H inductor, forming a low-loss power stage that ensures smooth current delivery and low output ripple. Multiple 220 μ F electrolytic capacitors are used for both input and output filtering, minimizing voltage fluctuations and enhancing system stability.

The fixed 5 V output is ideal for powering logic circuits, and sensors.

An on-board LED indicator provides a simple visual confirmation of power status, while the screw terminals (KF128-5.0-2P) allow secure and convenient wiring.

The module operates within a 7V to 25V input range, making it compatible with standard 12V and 24V systems.

Thanks to the LM2596's built-in thermal shutdown and short-circuit protection, the design offers dependable operation even under demanding load conditions.

Its compact PCB layout and straightforward design make it suitable for use in development setups, educational projects, or as a permanent power module in electronic assemblies.

Block Diagram

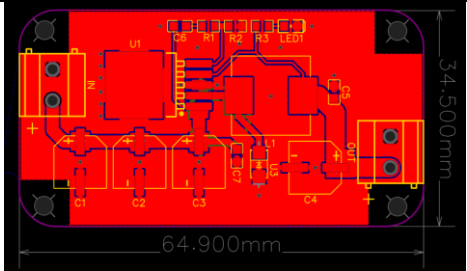
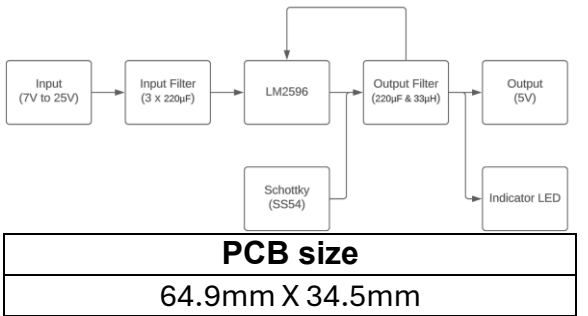


Figure 1. PCB Layout

Electrical Characteristics

Parameter	Min	Typ	Max	Unit
Input Voltage (VIN)	7	-	25	V
Output Voltage (VOUT)	4.9	5.0	5.1	V
Output Current	-	2	3	A
Efficiency	-	80	-	%
Ripple Voltage	-	30	50	mVpp

BOM (Bill of Materials)

Name	Designator	Quantity
220uF 35V	C1, C2, C3, C4	4
0.1uF 0805 X7R	C5, C6, C7	3
SS54	D1	1
KF128-5.0-2P	IN, OUT	2
33uH Dr125-330	L1	1
LED 0805 Red	LED1	1
47k 0805 1%	R1, R2	2
1k 0805 5%	R3	1
LM2596S-5.0	U1	1

Schematic

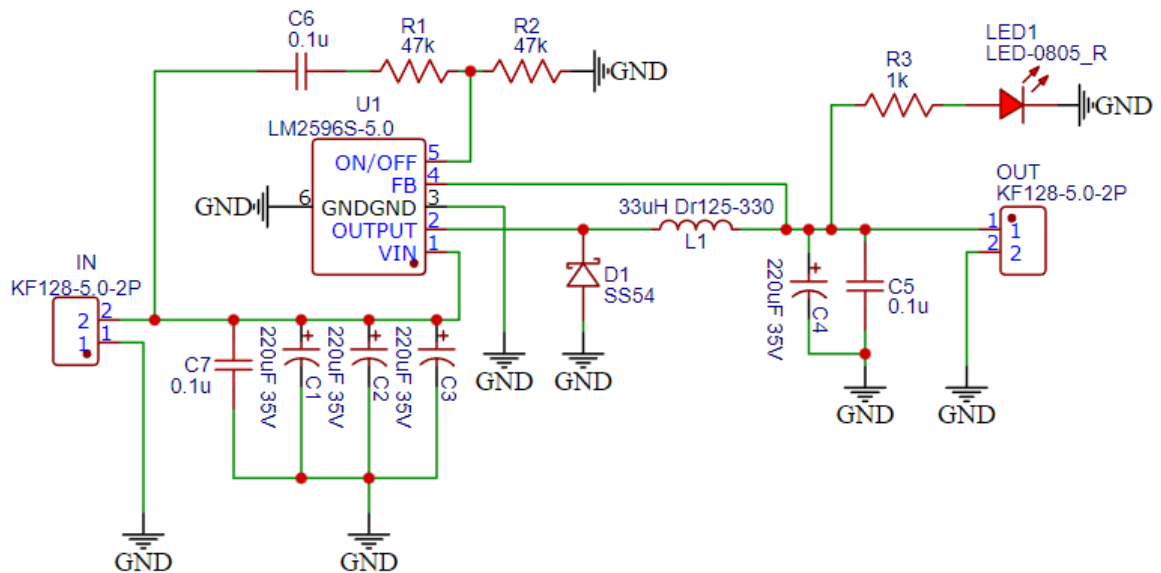


Figure 2. Schematic

3D image of the PCB

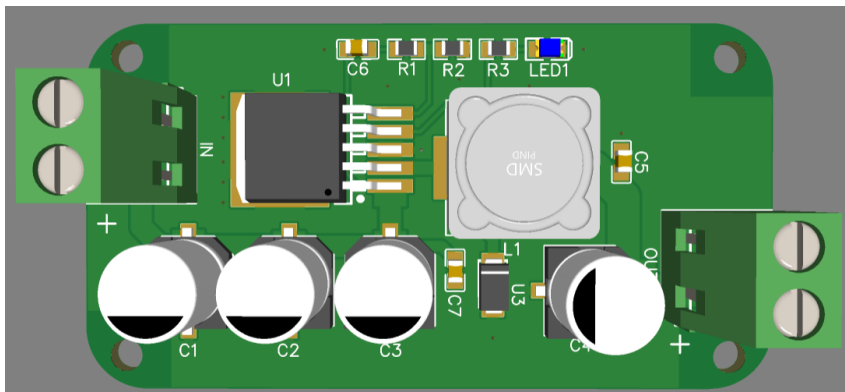


Figure 3. PCB Top Side

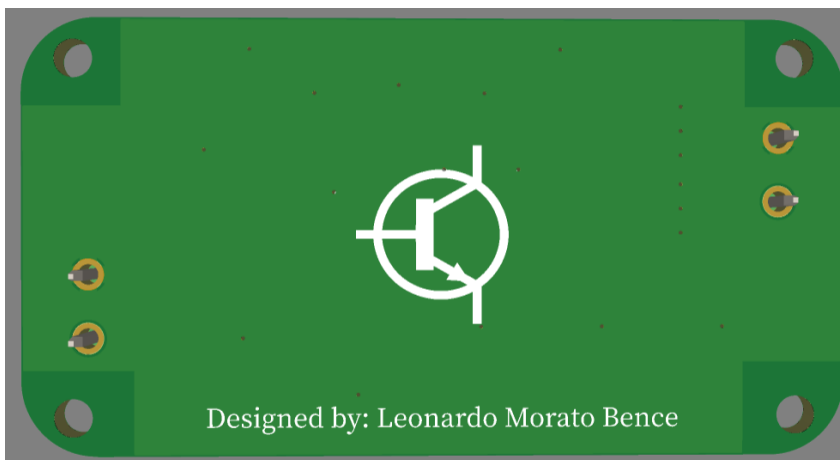


Figure 4. PCB Layout Bottom Side

PCB Assembled

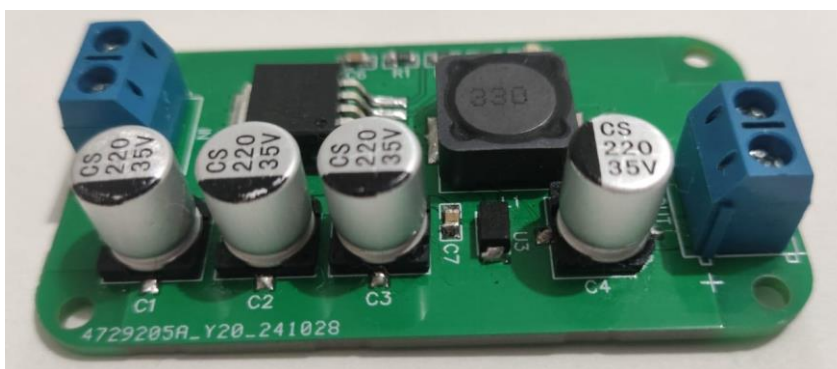


Figure 5. PCB Assembled

Disclaimer

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